

SymPy Bug Card

Bug 9: finite product masks an undefined factor

Task. Evaluate the finite product

$$\prod_{n=0}^2 \frac{\sin(\pi n)}{n-1}.$$

SymPy's answer. `product(sin( $\pi n$ )/( $n-1$ ), ( $n, 0, 2$ )) = 0`.

Correct answer. The product is nan, or should remain unevaluated.

Explanation. At $n = 1$, the factor is $\sin(\pi)/(1-1) = 0/0$, so the product contains an undefined term. Hence the finite product is not equal to 0, even though the factors at $n = 0$ and $n = 2$ vanish.

Likely root cause. In `Product.doit`, SymPy replaces the index by an integer-assumption dummy before evaluating the finite product, so $\sin(\pi n)$ collapses to 0 and masks the denominator singularity at $n = 1$. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-009-finite-product-masks-an-undefined-factor/](#)